Javier Emilio Alfonso Ramos

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Computational Chemist

Organic reactivity • DFT • Molecular Docking

Education

University of Havana Havana, Cuba
B.Sc. Chemistry 09/2017– 12/2022
Diplom 09/2019– 12/2022
Dissertation Project: "Origins of regioselectivy in 1,3-dipolar cycloaddition of acylnitrile ylides"
Supervisor: MSc. Gerardo M. Ojeda Carralero
Head of the Group: Prof. Luis A. Montero Cabrera

Experience

Research expertise

Organic reactivity: Reaction path modelling. Conceptual DFT. NCI. QM calculations

Bioinformatics: Molecular docking.

Experimental

Undergraduate experience in organic synthesis, analytical chemistry and elucidation of structures with several spectroscopic techniques (NMR, IR, UV, MS)

Language

Spanish: Native speaker

English: Good

Software

Quantum chemistry: Gaussian, ORCA, MultiWFN, MOPAC

Bioinformatics: AutoDock Vina, AutoDock4, OpenBabel

Graphics: VMD, PyMOL, CYLView, ChemDraw

Programming: Python, Bash, Matlab

Spectroscopy: MestReNova

HPC

Assembled and managed the cluster of the Faculty of Chemistry with CentOS, OpenHPC, Warewulf and Slurm.

Funding

Research grant PN223LH010-02X 2021–2023
from the National Program of Basic Sciences Republic of Cuba

Awards

Best Teaching Assistant Student of the Faculty of Chemistry, University of Havana (**2022**).

Best Student in Research of the Faculty of Chemistry, University of Havana (**2022**).

2nd place in the Prize Exam of Organic Chemistry I, Faculty of Chemistry, University of Havana (**2019**).

3rd place in the Organic Chemistry Commission of the Student's Scientific Day, Faculty of Chemistry, University of Havana (**2019**). Project: "Virtual Screening and synthesis of thiadiazine-2-thione-tetrazole hybrids".

1st place in the Bibliographic Review Commission of the Student's Scientific Day, Faculty of Chemistry, University of Havana (**2019**). Project: "Design of a research project: synthesis of novel thiadiazine-2-thione-tetrazole hybrids".

Special Prize of the Higher Education Ministry in the Student's Scientific Day, Faculty of Chemistry, University of Havana (**2018**). This prize is given to the junior student with the most outstanding research work.

3rd place in the Prize Exam of General Chemistry of the Faculty of Chemistry, University of Havana (**2017**).

Gold Medal in the **National Chemistry Olympiad**, Cuba (**2015**).

Gold Medal in the **National Chemistry Olympiad**, Cuba (**2014**).

Silver Medal in the **National Chemistry Olympiad**, Cuba (**2013**).

Teaching

Practical classes of *Spectroscopy* to 2nd-year students, Bachelor in Biochemistry, Faculty of Biology (2021-2022).

Practical classes of *Spectroscopy* to 4th-year students, Bachelor in Chemistry, Faculty of Chemistry (2022-2023).

Practical classes of *Informatics in Chemistry* to 1st-year students, Bachelor in Chemistry, Faculty of Chemistry (2021-2022).

Practical classes of *General Chemistry* to 1st-year students, Bachelor in Biochemistry, Faculty of Biology (2021-2022).

Workshops of *Fundamentals of Structure and Bonding* to 1st-year students, Bachelor in Chemistry, Faculty of Chemistry (2019-2022).

Practical classes of *General Chemistry* to 1st-year students, Bachelor in Microbiology, Faculty of Biology (2018-2019).

Independent Tutor of *General Chemistry* and *Organic Chemistry* to undergraduate student (2018-present)

Affiliation

Cuban Chemical Society 2017-

Posters & Conference Presentations

I International Symposium CIDEM 2022
Varadero, Matanzas, Cuba. Poster.

II LatinXChem 2021
Online Twitter event. Poster #Comp099.

XXIII Conference of Chemistry 2020
Santiago de Cuba, Cuba. Online poster.

IV Iberoamerican Symposium on Organic Chemistry 2019
Santa María Key, Villa Clara, Cuba. Poster.

XII Symposium on Molecular Design and Bioinformatics 2019
Santa María Key, Villa Clara, Cuba. Poster.

Student's Scientific Day 2019
Faculty of Chemistry, University of Havana. Commission of Research Reviews. Oral presentation.

Student's Scientific Day 2019
Faculty of Chemistry, University of Havana. Commission of Organic Chemistry. Oral presentation.

Student's Scientific Day 2018
Faculty of Chemistry, University of Havana. Commission of Research Reviews. Oral presentation.

Student's Scientific Day 2018
Faculty of Chemistry, University of Havana. Commission of Computational and Theoretical Chemistry. Oral presentation.

Publications

Articles

(1) Coro-Bermello, J.; López-Rodríguez, E.R.; **Alfonso-Ramos, J.E.**; Alonso, D.; Ojeda-Carralero, G.M.; Prado, G.A.; Moreno-Castillo; E. Identification of novel thiadiazin derivatives as potentially selective inhibitors towards trypanothione reductase from Trypanosoma cruzi by molecular docking using the numerical index poses ratio Pr and the binding mode analysis. *SN Appl. Sci.* 3, 376 (2021).

(2) Pillitteri, S.; Ranjan, P.; Ojeda-Carralero, G.M.; Vásquez, L.Y.; **Alfonso-Ramos, J.E.**; Van der Eycken, E.V.; Sharma, U. Merging dual photoredox/cobalt catalysis and boronic acid (derivatives) for the Minisci reaction in batch and continuous-flow. *Organic Chemistry Frontiers* (2022). Available in *ChemRvix.*

(3) **Alfonso-Ramos, J.E.**; Van Lommel, R.; Hernández-Castillo, D.; De Proft, F.; González-Alemán, R.; Van der Eycken, E.V.; Ojeda-Carralero, G.M. Origins of the regioselectivity in 1,3-dipolar cycloadditions of acyl-isocyanide ylides. *Eur. J. Org. Chem.* , (2022).

Profiles & Contact

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